

Sathish Kumar

Machine Learning Engineer | MLOps | GenAI & RAG

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PROFESSIONAL SUMMARY

Machine learning engineer with 6 years building AI systems that run in production, mostly on AWS and Azure. Recent work is largely LLM and RAG applications backed by Neo4j knowledge graphs, plus the MLOps that keeps them running: data pipelines, deployment, and watching for drift after launch. Comfortable taking a problem from raw data through to a deployed model, with inference latency down [X]% and cloud costs down [X]% along the way.

EXPERIENCE

AdvanceSoft International Inc - AI Engineer

Dec 2024 – Present

- Built a Neo4j knowledge graph from retail product and catalog data on Azure, giving analytics, ML features, and RAG applications one source to read from (around [X]M entities).
- Set up Azure Data Factory pipelines to move structured and semi-structured product data into Azure Data Lake and the analytics layers below it, on the order of [X] records a day.
- Wrote the Python and Databricks (Spark) jobs that normalize product attributes and clear out duplicates before they reach the graph, which lifted the entity match rate by [X]%.
- Loaded and queried the graph with the Neo4j Python driver and Cypher, and used Graph Data Science and embedding similarity to build ML features and the retrieval behind RAG search and Q&A.
- Kept the pipelines healthy in production with Azure Data Factory, Azure Monitor, and Application Insights.

Integer IT Solutions - Machine Learning Engineer

May 2024 – Dec 2024

- Built Python NLP pipelines to score sentiment, pull out entities, and classify intent in customer feedback, with Azure Data Factory handling the text cleanup first.
- Trained and shipped models with Azure ML and MLflow, ran inference in Docker on AKS (p95 under [X]ms), and added retraining and drift checks so accuracy did not slip unnoticed.
- Put the models behind FastAPI endpoints and fed their output into the Azure Synapse and Power BI dashboards the business team relied on.
- Wired up monitoring through Azure Monitor and Application Insights, and ran explainability checks for bias before anything sensitive went live.
- Worked with business and data teams on requirements and kept CI/CD moving on Azure DevOps and Git.

Southern Illinois University - Research Engineer

Jan 2022 – Dec 2023

- Built and compared predictive models in Scikit-learn, XGBoost, and PyTorch on behavioral and cognitive research data to estimate participant outcomes.
- Designed PyTorch deep-learning models for the structured and unstructured data, and ran computer-vision work with CNNs and YOLOv5 for object detection on roughly [X]K images.
- Tried federated learning to train across separate datasets without moving the data, which the project's privacy rules required.
- Did the preprocessing and feature engineering in Pandas to keep experiments repeatable, and stood up small Flask services to compare models as they ran.
- Packaged the workflows in Docker, GPU setups included, so they ran the same on lab and student machines, and made the figures (Matplotlib, Seaborn) for papers and talks.

DotIn Solutions - Software Engineer

Jan 2020 – Dec 2021

- Provisioned AWS infrastructure (EC2, RDS, S3, VPC, IAM, autoscaling) with Terraform, and used Ansible to keep configuration consistent across instances.
- Built the Jenkins CI/CD pipelines that tested, packaged, and deployed our containerized apps, with SonarQube and PyTest in the loop to catch issues early.
- Ran Dockerized services on EKS with Helm using blue-green and canary rollouts, and moved the Spark ETL into containers so it behaved the same in every environment.
- Set up S3 ingestion for logs and metrics and wrote Lambda functions to handle snapshots and lifecycle cleanup, so we stopped paying for storage we did not use.
- Stood up the monitoring stack (ELK, Grafana, CloudWatch) with SNS alerts, and helped fold early model-build steps into the pipelines.

SKILLS

Machine Learning & GenAI: PyTorch, Scikit-learn, XGBoost, Hugging Face Transformers, OpenCV, NLP, Computer Vision, LangChain, LangGraph, CrewAI, RAG, Pinecone, FAISS, Neo4j

MLOps & CI/CD: MLflow, Weights & Biases, Docker, Kubernetes, Airflow, dbt, Terraform, CI/CD, GitHub Actions, Jenkins

Cloud & Infrastructure: AWS (SageMaker, Bedrock, Glue, Redshift, Kinesis, EC2, S3, Lambda), Azure (ML, Data Factory, Synapse, AKS), GCP (Vertex AI, BigQuery), Databricks

Data Engineering: Python, PySpark, SQL, PL/SQL, PostgreSQL, MySQL, MongoDB, Snowflake, Kafka, FastAPI, Flask

Observability & Security: Prometheus, Grafana, CloudWatch, OpenTelemetry, IAM, OAuth2.0, Secrets Manager, RBAC

PROJECTS

Simulation of Autonomous Vehicles Using Federated Learning -

PyTorch, YOLOv5, Flower Framework, BDD100K

- Put together a federated-learning setup with Flower and YOLOv5 to mimic training object detection across edge devices, using the BDD100K driving dataset.
- Wrote the training and evaluation in Python so runs were repeatable and aggregation kept going even if a node dropped out.

Personality Prediction via Ensemble Learning for Real-Time Scoring -

Python, Flask, MongoDB, Scikit-learn, Docker

- Built a personality-prediction model with Random Forest and Gradient Boosting, served through a Dockerized Flask API (around [X]% accuracy).
- Stored predictions and history in MongoDB and retrained on user feedback as it came in.

EDUCATION

Southern Illinois University , M.S. Computer Science • GPA 3.83	Carbondale, IL • May 2024
Anurag Group of Institutions , B.Tech Computer Science • GPA 3.66	Hyderabad, India • May 2021